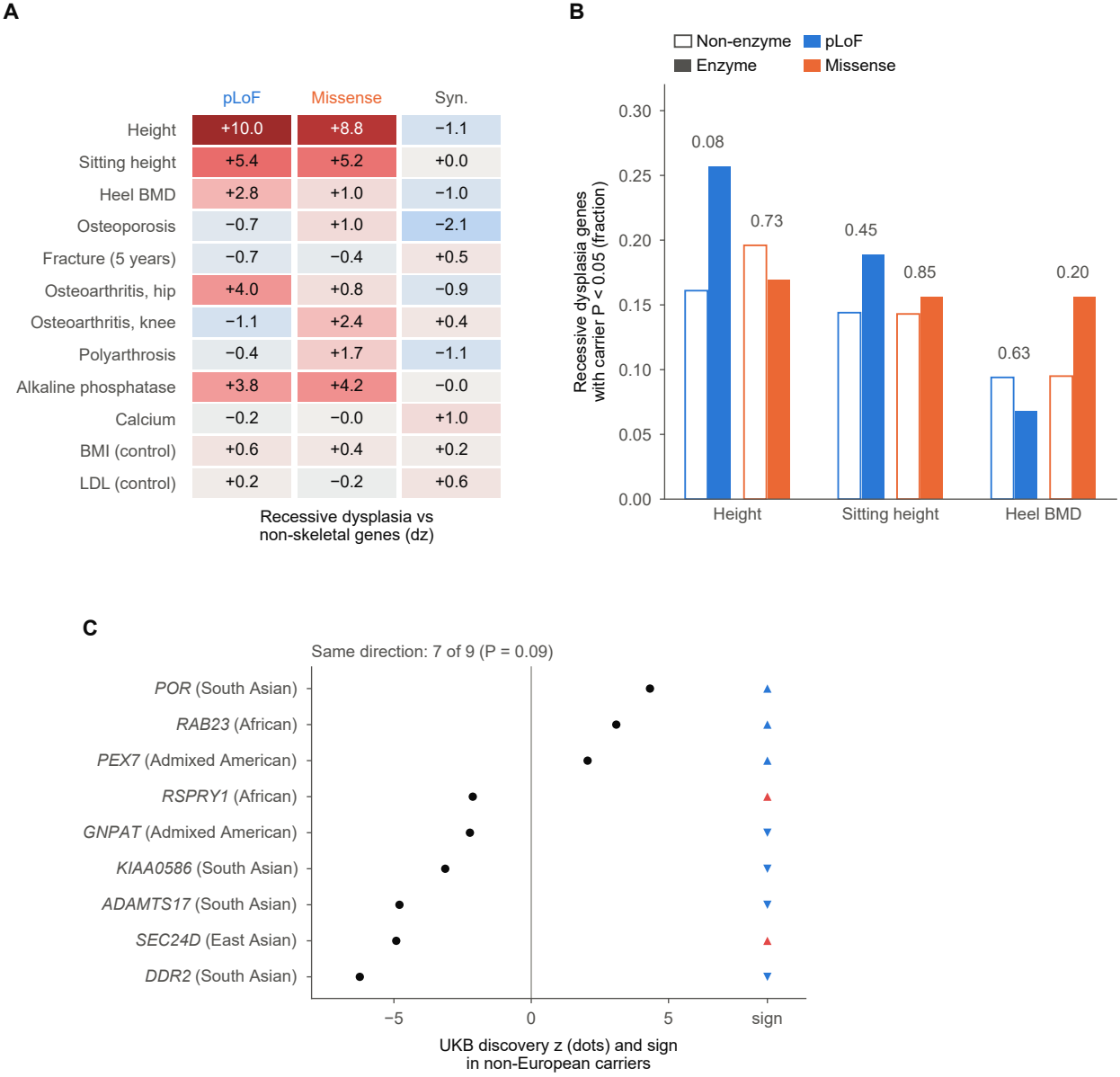

A
pLoF
Missense
Syn.
Height +10.0 +8.8 −1.1
Sitting height +5.4 +5.2 +0.0
Heel BMD +2.8 +1.0 −1.0
Osteoporosis −0.7 +1.0 −2.1
Fracture (5 years) −0.7 −0.4 +0.5
Osteoarthritis, hip +4.0 +0.8 −0.9
Osteoarthritis, knee −1.1 +2.4 +0.4
Polyarthrosis −0.4 +1.7 −1.1
Alkaline phosphatase +3.8 +4.2 −0.0
Calcium −0.2 −0.0 +1.0
BMI (control) +0.6 +0.4 +0.2
LDL (control) +0.2 −0.2 +0.6
Recessive dysplasia vs non-skeletal genes (dz)
B
Non-enzyme
Enzyme
pLoF
Missense
Recessive dysplasia genes with carrier P < 0.05 (fraction)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.08
0.73
0.45
0.85
0.63
0.20
Height
Sitting height
Heel BMD
C
Same direction: 7 of 9 (P = 0.09)
POR (South Asian)
RAB23 (African)
PEX7 (Admixed American)
RSPRY1 (African)
GNPAT (Admixed American)
KIAA0586 (South Asian)
ADAMTS17 (South Asian)
SEC24D (East Asian)
DDR2 (South Asian)
−5
0
5
sign
UKB discovery z (dots) and sign in non-European carriers